www.csvtuonline.com

337512 (37)

BE (5th Semester)

Examination, Nov-Dec., 2013

Branch : Mechanical

INTERNAL COMBUSTION ENGINES

Time Allowed : Three Hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : Solve all questions. In every question part 'a' is compulsory but solve any one of part 'b' or 'c'. Assume suitable data if required. Use of steam tables are permitted.

Q.1. (a) What is dual fuel engine. 2

www.csvtuonline.com

www.csvtuonline.com

www.csvtuonline.com

(b) Show that the compression ratio for the maximum work to be done per second per kilogram in an Otto engine cycle between upper and lower limits of absolute temperatures T_3 and T_1 is given by

$$R_c = \left(\frac{T_3}{T_1}\right)^{\frac{1}{2(\gamma-1)}}$$ 14

(c) The diameter and stroke of two stroke Diesel engine are 8 cm and 12 cm respectively. The clearance volume is 36.2 cm^3. The exhaust port closes after 45^0 crank rotation from BDC. The fuel valve opens at TDC and closes after 30^0 crank rotation from TDC. Find the air standard efficiency of the cycle.

www.csvtuonline.com

www.csvtuonline.com

www.csvtuonline.com

If the relative efficiency is 56% find the specific fuel consumption assuming the calorific value of the fuel used in 42000 kJ/kg. 14

Q.2. (a) What is structure of petroleum fuel ? 2

(b) What is fuel volatility and how does it affect the engine efficiency, volumetric efficiency, crank case dilution, carburation, and power output. 14

(c) Explain what is meant by detonation, knocking and preignition. What is antiknock rating of fuel and how it is used in deciding octane number and cetane number. 14

www.csvtuonline.com

www.csvtuonline.com

Q.3. (a) Define carburation. 2

(b) Describe a petrol injection system and mention its advantages and disadvantages. 14

(c) A simple carburettor under a certain condition delivers 5.45 kg/h of petrol with an air fuel ratio of 15 : 1. The fuel jet area is 2 mm^2 with a coefficient of discharge of 0.75. If the tip of the fuel jet is 0.635 cm above the level of petrol in the float chamber and the venturi coefficient is assumed to be 0.80, calculate

(i) Venturi throat diameter

www.csvtuonline.com

www.csvtuonline.com

(ii) Velocity of air across the venturi throat

Take density of air = 1.29 kg/m^3 and specific gravity of the fuel is 0.72. 14

Q.4. (a) Define atomization of fuel. 2

(b) Describe battery and magneto system by separate figures and make comparison of advantages and disadvantages. 14

(c) Explain the necessity of engine cooling. Describe the two main types of cooling systems. 14

Q.5. (a) What is performance of I.C. Engines ? 2

(b) Describe how the testing of the performance of I.C. engines are done for BHP, I.H.P, air and fuel consumption, load and speed. 14

3

www.csvtuonline.com

www.csvtuonline.com

www.csvtuonline.com

www.csvtuonline.com

(c) A petrol engine uses a fuel of calorific value 43963.5 kJ/kg. The compression and expansion curves follow the law $pv^{1.33}$ = constant. At 25% and 75% of compression stroke, the pressures are 1.96 bar and 5.4 bar. The relative and mechanical efficiencies of engine are 48% and 78% respectively. Find specific fuel consumption based on brake power. 14

www.csvtuonline.com

www.csvtuonline.com